

Soumojit Chowdhury

Lead Data Scientist (Assistant Director) | AML & Fraud Analytics | Financial Crime Prevention | Money Mule Detection
Kolkata, India • +91-9874113384 • soumojit.chowdhury@gmail.com • linkedin.com/in/soumojit-chowdhury

PROFESSIONAL SUMMARY

Lead Data Scientist (Assistant Director) with 10+ years of ML experience specialising in Anti-Money Laundering (AML), fraud analytics, financial crime prevention, and real-time money mule detection. Part of HSBC's Global Fraud Analytics function as one of three RTMM programme leads across Australia, Singapore, Malaysia, and India — screening ~6–7 million customers daily. Deep expertise in ML model development, graph analytics (TigerGraph), behavioural biometrics (BioCatch), real-time transaction monitoring (Feedzai), and GenAI/LLM applications for financial crime operations. Proven FLOD track record: RTMM controls 40–50% of market money mule share at one-quarter Second Line's investigator cost, with ~40% end-to-end SAR conversion. Regular engagement with MAS, FIU India, AUSTRAC, and senior leadership at VP/SVP level.

CORE COMPETENCIES

Anti-Money Laundering (AML), Fraud Analytics, Money Mule Detection, Financial Crime Prevention, Transaction Monitoring SAR & UCM Filing Strategy, Graph Analytics (TigerGraph), Feedzai Rule Engineering, BioCatch Behavioural Biometrics, Large Language Models (LLMs)

Generative AI & Prompt Engineering, Model Monitoring & Validation, Model Governance & Risk Management, Regulatory Engagement (MAS / RBI / AUSTRAC), Python (Advanced) / PySpark / BigQuery

PROFESSIONAL EXPERIENCE

HSBC – Lead Data Scientist (Assistant Director, Data & Analytics)

Mar 2022 – Present

Global Fraud Analytics | IWPB Financial Crime Prevention | RTMM Analytics | Markets: Australia • Singapore • Malaysia • India

- **Programme Lead — RTMM Analytics:** One of three analytics leads within Global Fraud Analytics responsible for end-to-end money mule detection strategy, model and rule development, and programme delivery across four markets — generating ~2,600 monthly alerts (~1,000 AUS, ~600 IND, ~600 SGP, ~400 MYS) and screening approximately 6–7 million customers daily.
- **First Line of Defence Efficiency:** Elevated alert precision from 10–20% to ~50% through rule engineering, model recalibration, and feature development. In mature markets, RTMM controls 40–50% of total money mule-suspected customer share using roughly one-quarter of Second Line Compliance's investigator headcount — a highly cost-efficient financial crime prevention control.
- **SAR & Financial Intelligence Outcomes:** Achieved ~40% end-to-end SAR conversion on RTMM alerts. Programme feeds Unusual Cash Movement (UCM) filings to Compliance (Second Line), which files Suspicious Activity Reports (SARs) with MAS, FIU India, and AUSTRAC.
- **Early Detection Modelling — New-to-Bank Mules:** Developed and deployed an Isolation Forest-based early detection model for New-to-Bank money mule identification, achieving ~60% precision (UCM filing rate). Implemented model monitoring, model validation, population stability analysis, base rate shift detection, and threshold recalibration across the mule detection portfolio.
- **Graph Analytics — TigerGraph:** Designed a TigerGraph network analytics pipeline for mule ring detection using PageRank, degree centrality, and community detection across a financial crime entity schema to surface connected mule networks from escalated cases.
- **Real-Time Transaction Monitoring — Feedzai:** Architecting Feedzai transaction monitoring deployment (production Q3 2025); designing queue architecture, PySpark feature maps, rule engineering framework, and treatment strategies. Roadmap includes supervised transactional ML models for money mule detection in subsequent releases.
- **Behavioural Biometrics — BioCatch:** Integrated BioCatch behavioural risk signals into the mule scoring framework; built a compound scoring architecture to address detection gaps in the Australian market, improving risk coverage and investigator decision quality.
- **GenAI & LLM — Investigation Assist Prototype:** Built a GenAI-powered Investigation Assist tool using LLMs for financial crime case summarisation and UCM filing guidance to streamline investigator workflows. Submitted for AI governance approval; incorporated graph analytics context, behavioural signals, and multi-source risk indicators across all four markets.

- **Feature Engineering & ML Infrastructure:** Engineered a multi-market ML feature library in BigQuery (~200 features) across customer, account, transaction, and digital footprint dimensions; built a modular Python (Advanced)/BigQuery codebase for scalable cross-market deployment.
- **Regulatory & Stakeholder Engagement:** Presents to VPs, SVPs, and regulators (MAS, FIU India, AUSTRAC) on model performance, typology coverage, and financial crime strategy. Authors programme analytics reporting and model governance documentation. Manages and mentors 4 junior analysts within MRM compliance frameworks.

Tiger Analytics – Data Scientist

Mar 2021 – Mar 2022

- **Industrial IoT & Predictive Analytics:** Built real-time anomaly detection and time-series forecasting models (LSTM, Prophet) on sensor telemetry for a Fortune 500 steel manufacturer via Apache Spark pipelines, enabling ~\$50M in cost avoidance and ~\$5M in energy savings through proactive failure detection and production optimisation.

Cognizant – Associate Data Scientist

Aug 2018 – Mar 2021

- **Insurance Marketing Analytics:** Built propensity scoring and segmentation models (K-Means, DBSCAN) for insurance campaign optimisation, improving lead conversion by 14%. Delivered model insights to non-technical senior leadership.

Infosys – Analyst

Aug 2015 – Aug 2018

- **Fraud Detection:** Developed ML models for energy meter tampering and utility fraud detection for a major US utility client, reducing manual investigation time by 60%.

EDUCATION

B.Tech, Computer Science & Engineering | West Bengal University of Technology | 2011 – 2015

DOMAIN EXPERTISE

Financial Crime: Anti-Money Laundering (AML), Fraud Analytics, Money Mule Detection, Financial Crime Prevention, Transaction Monitoring, SAR & UCM Filing, STR, KYC, Enhanced Due Diligence (EDD), First Line of Defence (FLOD), Typology Analysis, Financial Crime Risk Strategy

Regulatory: MAS (Singapore), RBI (India), AUSTRAC (Australia), FIU India, Model Risk Management (MRM), Model Governance, Compliance Liaison

TECHNICAL SKILLS

Platforms & Vendors: Feedzai (Rule Engineering, Queue Architecture, PySpark), TigerGraph (Graph Analytics), BioCatch (Behavioural Biometrics), Azure ML Studio, Azure Data Factory, GCP Vertex AI, MLflow, Airflow

ML, AI & GenAI: Isolation Forest, Transformer Encoder, LSTM, XGBoost, LightGBM, Scikit-learn, TensorFlow, PyTorch, BERT, LLMs, Generative AI, Prompt Engineering, Anomaly Detection, SHAP Explainability, Model Monitoring, Model Validation, PSI, Drift Detection, A/B Testing

Big Data & Engineering: Python (Advanced), PySpark, BigQuery, SQL, Apache Spark, Pandas, NumPy, Jupyter Notebooks, SAS (modernising to Altair), Git

BI & Visualisation: Looker, Qlik Sense